Chi Lee

ADDITIVE MANUFACTURING · PARAMETRIC DESIGN · PROTOTYPING

Barcelona, Spain · +34 665 793 655 · fu6fu62@gmail.com
linkedin.com/in/chi-l-589166229 · Portfolio: leeeeeeeeeerichileeeeeeeeee.github.io/portfolio

Architectural designer and researcher translating material and geometric requirements into fabricable components. Project experience spans parametric Rhino/Grasshopper modelling, full-scale PETG FDM prototypes and ABB robotic earth injection. Combines design iteration, natural-material experimentation and physical fabrication with architectural practice.

RELEVANT SKILLS

Design	Rhino, Grasshopper; parametric component design; AutoCAD, SketchUp
Fabrication	FDM 3D printing (PETG); ABB / UR fabrication workflows; prototype iteration; CNC-based jigs
Sensing	LiDAR point-cloud processing; drone photogrammetry (Metashape)
Coursework	Python, ROS/MoveIt; introductory PyTorch / TensorFlow exposure

SELECTED PROJECT EXPERIENCE

2025–2026

SCULPTA | Parametric PETG components and full-scale FDM prototypes

Individual master's thesis · MADE, ETSAV–UPC

- Translated microalgal cultivation requirements into geometric rules and a parametric Rhino/Grasshopper model for a closed PETG reactor unit.
- Produced full-scale FDM specimens and revised the design after handling failures: increased nominal wall thickness from 1.0 to 1.2 mm and smoothed the shoulder geometry.
- Designed replaceable unit groups, independently valved circuits and a 600 mm maintenance zone. Watertightness, cultivation and thermal performance were proposed as subsequent tests, not validated outcomes.

2024–2025

Terraweave | Robotic earth injection into woven willow scaffolds

Team project · IAAC MRAC · with Elizabeth Frias and Krystyne Kontos

- Contributed to design development, Grasshopper-based robot control and toolpath programming, prototype fabrication and project video production.
- Worked within an ABB robotic fabrication workflow combining natural-fibre scaffolds with clay/earth injection.
- The team also developed computer-vision-assisted injection-point identification.

blog.iaac.net/terraweave/

2025

Robotic Scanning for Building Anomaly Detection

Team project · IAAC MRAC · with Elizabeth Frias, Lauren Deming and Krystyne Kontos

- Operated the robot, processed LiDAR point clouds and analysed roof anomalies.
- The team used Husky A200 LiDAR scanning and compared the approach with drone photogrammetry in Metashape.
- Used point-cloud geometry to locate roof openings and estimate repair-material quantities without direct physical roof access.

blog.iaac.net/robotics-solutions-for-building-anomaly-detection/

2024–2025 **Selected IAAC fabrication coursework and team projects**

Robotics and Advanced Construction I (MRAC)

- Scanning for Reconstruction: team workflow linking UR10e depth-camera scanning to mesh-based, non-planar 3D-printing toolpaths using Grasshopper, ROS and Python.
- Cork ReForm: robotic 3D printing with a cork-biomass mix. Breathing Curtain Wall: exploration of woven willow facade modules.

PROFESSIONAL EXPERIENCE

2026–present **Business Development Manager | Ourbit Exchange**

Remote · Since June 2026

- Source and qualify international partners, crypto creators and trading communities; conduct partnership and affiliate discussions and report pipeline progress.

2023–2024 **Interior Designer | Changzhou Wuji Architectural Space Design**

China

- Design team member on “Mochun”, winner of the 2023 Jintang Award for Best Residential Apartment Design.

2023 **Interior and Architectural Designer | Changzhou Jingling Architect**

China

- Contributed to architectural and road-landscape planning for the northern area of Changzhou Economic Development Zone.

2020–2022 **Architectural Designer | Daan TMA Architect & Associates**

Project Department · Taipei, Taiwan

- Worked on building planning, facade design and design competitions for residential, high-rise and commercial projects.

Earlier Web3 experience includes AI video and KOL content strategy at MEXC, and blockchain community marketing at Gamania (GASH).

EDUCATION

2025–2026 **Master’s Degree in Architectural Design Ecology in the Digital Age (MADE)**

ETSAV, Universitat Politècnica de Catalunya (UPC) · Completed September 2026 · Taught in English

- Thesis: A “Breathing” Façade: SCULPTA, a 3D-Printed PETG Microalgal Reactor. Supervisors: David López López and Marta Domènech Rodríguez.

2024–2025 **Robotics and Advanced Construction I (MRAC)**

Institute for Advanced Architecture of Catalonia (IAAC), Barcelona

2015–2020 **Bachelor of Architecture**

Feng Chia University, Taichung, Taiwan

SELECTED PUBLICATIONS

2026 **Material-driven formation of architectural character in Huangshi sandstone masonry**

Shi, B., Chao, Z., Lee, C., et al. · npj Heritage Science · Co-author

doi.org/10.1038/s40494-026-02897-7

2026 **A 3D-based diagnostic workflow for conservation decision-making of traditional timber architecture**

Built Heritage · Co-author · Accepted for publication

LANGUAGES

Mandarin Chinese (native) · English (professional working proficiency; MSc taught in English) · Spanish (A1) · French (A1)